

Práctica S4: Rompiendo las Cadenas del Host - Aislamiento Manual de Procesos - Jorge Parra ITIT 13104

vagrant up y ssh

```
Administrator: Windows PowerShell
default: shared folder errors, please make sure the guest additions within the
default: virtual machine match the version of VirtualBox you have installed on
default: your host and reload your VM.
default:
default: Guest Additions Version: 6.1.50
default: VirtualBox Version: 7.2
==> default: Configuring and enabling network interfaces...
==> default: Mounting shared folders...
default: C:/Users/jorge/OneDrive/Escritorio/9th-semester-ulsal/Toledo - Instrumentacion virtual para moviles/Parcial_1/practica-s3 => /vagrant
--> default: Running provisioner: shell...
default: Running: inline script
default: Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
default: Get:2 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
default: Get:3 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
default: Get:4 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
default: Get:5 http://archive.ubuntu.com/ubuntu focal/universe amd64 Packages [8628 kB]
default: Get:6 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3611 kB]
default: Get:7 http://archive.ubuntu.com/ubuntu focal/universe Translation-en [5124 kB]
default: Get:8 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [518 kB]
default: Get:9 http://archive.ubuntu.com/ubuntu focal/universe amd64 c-n-f Metadata [265 kB]
default: Get:10 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3828 kB]
default: Get:11 http://archive.ubuntu.com/ubuntu focal/multiverse amd64 Packages [144 kB]
default: Get:12 http://archive.ubuntu.com/ubuntu focal/multiverse Translation-en [104 kB]
default: Get:13 http://archive.ubuntu.com/ubuntu focal/multiverse amd64 c-n-f Metadata [9136 B]
default: Get:14 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [4004 kB]
default: Get:15 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1052 kB]
default: Get:16 http://security.ubuntu.com/ubuntu focal-security/universe Translation-en [220 kB]
default: Get:17 http://security.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [22.4 kB]
default: Get:18 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [26.7 kB]
default: Get:19 http://security.ubuntu.com/ubuntu focal-security/multiverse Translation-en [6416 B]
default: Get:20 http://archive.ubuntu.com/ubuntu focal-updates/main Translation-en [601 kB]
default: Get:21 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3984 kB]
default: Get:22 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [604 B]
default: Get:23 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1275 kB]
default: Get:24 http://archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [302 kB]
default: Get:25 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [29.3 kB]
default: Get:26 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [29.8 kB]
default: Get:27 http://archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [8284 B]
default: Get:28 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [688 B]
default: Get:29 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [46.1 kB]
default: Get:30 http://archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16.2 kB]
default: Get:31 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1420 B]
default: Get:32 http://archive.ubuntu.com/ubuntu focal-backports/restricted amd64 c-n-f Metadata [116 B]
default: Get:33 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25.0 kB]
default: Get:34 http://archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]
default: Get:35 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
default: Get:36 http://archive.ubuntu.com/ubuntu focal-backports/multiverse amd64 c-n-f Metadata [116 B]
default: Fetched 34.3 MB in 12s (2785 kB/s)
default: Reading package lists...
default: Reading package lists...
default: Building dependency tree...
default: Reading state information...
default: curl is already the newest version (7.68.0-1ubuntu2.25).
default: curl set to manually installed.
default: 0 upgraded, 0 newly installed, 0 to remove and 23 not upgraded.
PS C:\Users\jorge\OneDrive\Escritorio\9th-semester-ulsal\Toledo - Instrumentacion virtual para moviles\Parcial_1\practica-s3> _
```

```

PS C:\Users\jorge\OneDrive\Escritorio\9th-semester-ulsa\Toledo - Instrumentacion virtual para moviles\Parcial_1\practica-s3> vagrant ssh
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep  6 02:57:18 UTC 2026

System load:          0.06
Usage of /:            4.2% of 38.70GB
Memory usage:         11%
Swap usage:           0%
Processes:            128
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:1a:a2ff:fe1e:9db7

Expanded Security Maintenance for Infrastructure is not enabled.

21 updates can be applied immediately.
18 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

147 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

vagrant@ubuntu-focal:~$ echo "ID: 13104"
ID: 13104
vagrant@ubuntu-focal:~$

```

Hito 1 - Aislamiento del proceso (host y bunker)

```

root@ubuntu-focal: /home/vagrant
vagrant@ubuntu-focal:~$ echo "ID: 13104"
ID: 13104
vagrant@ubuntu-focal:~$ sudo unshare --fork --pid --mount-proc bash
root@ubuntu-focal: /home/vagrant# echo "ID: 13104"
ID: 13104
root@ubuntu-focal: /home/vagrant# ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1   8960  3908 pts/0    S    03:00   0:00 bash
root           8  0.0  0.1  10612  3376 pts/0    R+   03:01   0:00 ps aux
root@ubuntu-focal: /home/vagrant# echo "ID: 13104"
ID: 13104
root@ubuntu-focal: /home/vagrant# readlink /proc/self/ns/pid
pid:[4026532190]
root@ubuntu-focal: /home/vagrant#

```

Hito 2 - El Espejismo del Árbol de Procesos

```

vagrant@ubuntu-focal: ~
pid:[4026532190]
root@ubuntu-focal:/home/vagrant# exit
exit
vagrant@ubuntu-focal:~$ echo "ID: 13104"
ID: 13104
vagrant@ubuntu-focal:~$ ps aux

```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.6	0.6	103804	12812	?	Ss	02:52	0:05	/sbin/init
root	2	0.0	0.0	0	0	?	S	02:52	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	I<	02:52	0:00	[rcu_gp]
root	4	0.0	0.0	0	0	?	I<	02:52	0:00	[rcu_par_gp]
root	6	0.0	0.0	0	0	?	I<	02:52	0:00	[kworker/0:0H-kblockd]
root	8	0.0	0.0	0	0	?	I<	02:52	0:00	[mm_percpu_wq]
root	9	0.0	0.0	0	0	?	S	02:52	0:00	[ksoftirqd/0]
root	10	0.0	0.0	0	0	?	I	02:52	0:00	[rcu_sched]
root	11	0.0	0.0	0	0	?	S	02:52	0:00	[migration/0]
root	12	0.0	0.0	0	0	?	S	02:52	0:00	[idle_inject/0]
root	14	0.0	0.0	0	0	?	S	02:52	0:00	[cpuhp/0]
root	15	0.0	0.0	0	0	?	S	02:52	0:00	[cpuhp/1]
root	16	0.0	0.0	0	0	?	S	02:52	0:00	[idle_inject/1]
root	17	0.0	0.0	0	0	?	S	02:52	0:00	[migration/1]
root	18	0.1	0.0	0	0	?	S	02:52	0:01	[ksoftirqd/1]
root	20	0.0	0.0	0	0	?	I<	02:52	0:00	[kworker/1:0H-kblockd]
root	21	0.0	0.0	0	0	?	S	02:52	0:00	[kdevtmpfs]
root	22	0.0	0.0	0	0	?	I<	02:52	0:00	[netns]
root	23	0.0	0.0	0	0	?	S	02:52	0:00	[rcu_tasks_kthre]
root	24	0.0	0.0	0	0	?	S	02:52	0:00	[kauditd]
root	25	0.0	0.0	0	0	?	I	02:52	0:00	[kworker/1:1-rcu_gp]
root	26	0.0	0.0	0	0	?	S	02:52	0:00	[khungtaskd]
root	27	0.0	0.0	0	0	?	S	02:52	0:00	[oom_reaper]
root	28	0.0	0.0	0	0	?	I<	02:52	0:00	[writeback]
root	29	0.0	0.0	0	0	?	S	02:52	0:00	[kcompactd0]
root	30	0.0	0.0	0	0	?	SN	02:52	0:00	[ksmd]
root	31	0.0	0.0	0	0	?	SN	02:52	0:00	[khugepaged]
root	77	0.0	0.0	0	0	?	I<	02:52	0:00	[kintegrityd]
root	78	0.0	0.0	0	0	?	I<	02:52	0:00	[kblockd]
root	79	0.0	0.0	0	0	?	I<	02:52	0:00	[blkcg_punt_bio]
root	80	0.0	0.0	0	0	?	I<	02:52	0:00	[tpm_dev_wq]
root	81	0.0	0.0	0	0	?	I<	02:52	0:00	[ata_sff]
root	82	0.0	0.0	0	0	?	I<	02:52	0:00	[md]
root	83	0.0	0.0	0	0	?	I<	02:52	0:00	[edac-poller]
root	84	0.0	0.0	0	0	?	I<	02:52	0:00	[devfreq_wq]
root	85	0.0	0.0	0	0	?	S	02:52	0:00	[watchdogd]
root	88	0.0	0.0	0	0	?	S	02:52	0:00	[kswapd0]
root	89	0.0	0.0	0	0	?	S	02:52	0:00	[ecryptfs-kthrea]
root	91	0.0	0.0	0	0	?	I<	02:52	0:00	[kthrotld]
root	92	0.0	0.0	0	0	?	I<	02:52	0:00	[acpi_thermal_pm]
root	93	0.0	0.0	0	0	?	S	02:52	0:00	[scsi_eh_0]
root	94	0.0	0.0	0	0	?	I<	02:52	0:00	[scsi_tmf_0]
root	95	0.0	0.0	0	0	?	S	02:52	0:00	[scsi_eh_1]
root	96	0.0	0.0	0	0	?	I<	02:52	0:00	[scsi_tmf_1]
root	98	0.0	0.0	0	0	?	I<	02:52	0:00	[vfio-irqfd-clea]
root	99	0.0	0.0	0	0	?	I<	02:52	0:00	[ipv6_addrconf]
root	108	0.0	0.0	0	0	?	I<	02:52	0:00	[kstrp]
root	111	0.0	0.0	0	0	?	I<	02:52	0:00	[kworker/u5:0]
root	124	0.0	0.0	0	0	?	I<	02:52	0:00	[charger_manager]
root	164	0.0	0.0	0	0	?	I<	02:52	0:00	[ttm_swap]
root	179	0.0	0.0	0	0	?	I<	02:52	0:00	[cryptd]
root	182	0.1	0.0	0	0	?	I	02:52	0:01	[kworker/1:2-mpt_poll_0]
root	184	0.0	0.0	0	0	?	I<	02:52	0:00	[mpt_poll_0]
root	185	0.0	0.0	0	0	?	I<	02:52	0:00	[mpt/0]
root	202	0.0	0.0	0	0	?	S	02:52	0:00	[scsi_eh_2]
root	203	0.0	0.0	0	0	?	I<	02:52	0:00	[scsi_tmf_2]

```
vagrant@ubuntu-focal: ~  
vagrant@ubuntu-focal:~$ echo "ID: 13104"  
ID: 13104  
vagrant@ubuntu-focal:~$ readlink /proc/self/ns/pid  
pid:[4026531836]  
vagrant@ubuntu-focal:~$
```

Hito 3 - Mapeo de Identidades Fracturadas

```
vagrant@ubuntu-focal: ~  
vagrant@ubuntu-focal:~$ echo "ID: 13104"  
ID: 13104  
vagrant@ubuntu-focal:~$ readlink /proc/self/ns/pid  
pid:[4026531836]  
vagrant@ubuntu-focal:~$ echo "ID: 13104"  
ID: 13104  
vagrant@ubuntu-focal:~$ readlink /proc/self/ns/pid  
pid:[4026531836]  
vagrant@ubuntu-focal:~$ echo "ID: 13104"  
ID: 13104  
vagrant@ubuntu-focal:~$ readlink /proc/self/ns/pid  
pid:[4026531836]  
vagrant@ubuntu-focal:~$
```

Hito 4 - El Límite del Metal

```

Select vagrant@ubuntu-focal: /sys/fs/cgroup/memory/resistencia
vagrant@ubuntu-focal:~$ mount | grep cgroup
tmpfs on /sys/fs/cgroup type tmpfs (ro,nosuid,nodev,noexec,mode=755)
cgroup2 on /sys/fs/cgroup/unified type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
cgroup on /sys/fs/cgroup/systemd type cgroup (rw,nosuid,nodev,noexec,relatime,xattr,name=systemd)
cgroup on /sys/fs/cgroup/freezer type cgroup (rw,nosuid,nodev,noexec,relatime,freezer)
cgroup on /sys/fs/cgroup/rdma type cgroup (rw,nosuid,nodev,noexec,relatime,rdma)
cgroup on /sys/fs/cgroup/net_cls,net_prio type cgroup (rw,nosuid,nodev,noexec,relatime,net_cls,net_prio)
cgroup on /sys/fs/cgroup/blkio type cgroup (rw,nosuid,nodev,noexec,relatime,blkio)
cgroup on /sys/fs/cgroup/pids type cgroup (rw,nosuid,nodev,noexec,relatime,pids)
cgroup on /sys/fs/cgroup/cpu,cpuacct type cgroup (rw,nosuid,nodev,noexec,relatime,cpu,cpuacct)
cgroup on /sys/fs/cgroup/perf_event type cgroup (rw,nosuid,nodev,noexec,relatime,perf_event)
cgroup on /sys/fs/cgroup/devices type cgroup (rw,nosuid,nodev,noexec,relatime,devices)
cgroup on /sys/fs/cgroup/memory type cgroup (rw,nosuid,nodev,noexec,relatime,memory)
cgroup on /sys/fs/cgroup/hugetlb type cgroup (rw,nosuid,nodev,noexec,relatime,hugetlb)
cgroup on /sys/fs/cgroup/cpuset type cgroup (rw,nosuid,nodev,noexec,relatime,cpuset)
vagrant@ubuntu-focal:~$ cd /sys/fs/cgroup/memory
vagrant@ubuntu-focal:/sys/fs/cgroup/memory$ sudo mkdir resistencia
vagrant@ubuntu-focal:/sys/fs/cgroup/memory$ cd resistencia
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ echo "ID: 13104"
ID: 13104
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ echo 52428800 | sudo tee memory.limit_in_bytes
52428800
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ echo "ID: 13104"
ID: 13104
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ echo $$ | sudo tee cgroup.procs
2998
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ echo "ID: 13104"
ID: 13104
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ dd if=/dev/zero of=/dev/null bs=100M count=1
Killed
vagrant@ubuntu-focal:/sys/fs/cgroup/memory/resistencia$ _

```

Explicacion

Hito 1: La Forja de la Burbuja

Se ejecutó el comando `unshare` con las opciones `--fork`, `--pid` y `--mount-proc` para crear un nuevo espacio de nombres de procesos. Esto permitió iniciar un entorno aislado donde el proceso `bash` aparece como PID 1.

Hito 2: El Espejismo del Árbol de Procesos

Se utilizó `ps aux` dentro y fuera del entorno aislado para comparar los procesos visibles. Dentro del búnker solamente se observaron los procesos pertenecientes al nuevo namespace, mientras que fuera se pudieron observar los procesos del sistema completo.

Hito 3: Mapeo de Identidades Fracturadas

Se utilizó `readlink /proc/self/ns/pid` para obtener el identificador del namespace de procesos. Al comparar el identificador dentro y fuera del búnker, se comprobó que ambos pertenecen a namespaces PID diferentes.

Hito 4: El Límite del Metal

Se creó un cgroup de memoria llamado `resistencia` y se estableció un límite de 50 MB mediante `memory.limit_in_bytes`. Después se agregó el proceso al cgroup y se ejecutó `dd` intentando consumir 100 MB, provocando que el kernel terminara el proceso al superar el límite de memoria establecido.